Lahore University of Management Sciences
School of Science and Engineering, Department of Electrical Engineering

AI600 – Deep Learning

Assignment 2: Quick, Draw! Challenge

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Programming Components (Public GitHub Repository):
https://github.com/abdull2325/ai600-pa2-25280050

This report documents architecture search experiments under assignment constraints: no CNNs, parameter budget $\leq$ 3,000,000, and training budget $\leq$ 40 epochs.

Problem Statement and Constraints

This assignment uses a 15-class subset of the Quick, Draw! dataset with input bitmaps flattened to 784-dimensional vectors. The objective is to classify doodles using **only MLP architectures** while respecting:

- No convolutional layers.
- Maximum trainable parameters: 3,000,000.
- Maximum training epochs: 40.

Data and Training Setup

- Training set: 60,000 labeled samples.
- Validation split: 80/20 (48,000 train, 12,000 validation).
- Test set: 15,000 unlabeled samples.
- Preprocessing: pixel normalization from [0, 255] to [0, 1].
- Loss: Cross-Entropy.
- Optimizer: AdamW with model-specific learning rates and weight decay.
- Regularization strategies explored: dropout, batch normalization, early stopping.

Part A/B/C: Architecture Search Results

Three models were trained as required:

- **Pancake (width-focused):** one wide hidden layer.
- **Tower (depth-focused):** six hidden layers with narrow width.
- **Champion:** tuned depth-width balance for best validation performance.

Model Configuration Summary

Model	Hidden Layers / Width	Params	Used Epochs	Best Val Acc
Pancake	[2048]	1,638,415	24	77.35%
Tower	[256,256,256,256,256,256]	536,847	28	78.21%
Champion	[1024,768,512,256]	2,125,071	34	79.16%

Table 1: Performance and efficiency comparison of the three required models.

Learning Curves

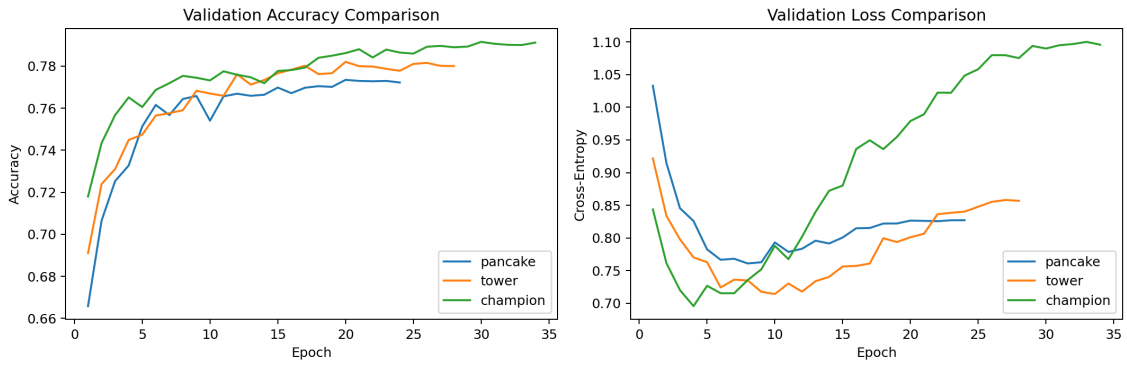
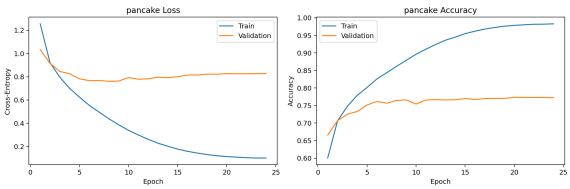
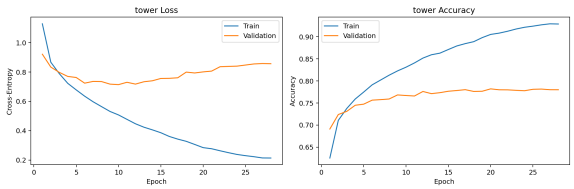


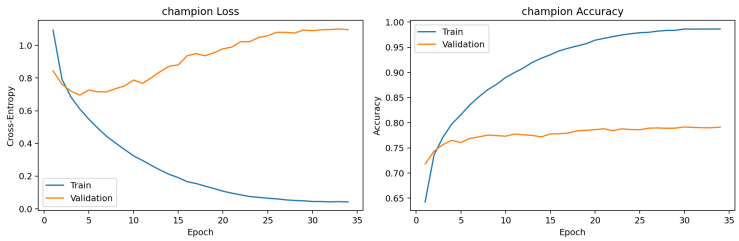
Figure 1: Validation accuracy and validation loss comparison across Pancake, Tower, and Champion models.



(a) Pancake curves



(b) Tower curves



(c) Champion curves

Figure 2: Training/validation loss and accuracy curves for all three models.

Part D: Theoretical Analysis

D1. Width vs Depth

The Universal Approximation Theorem states that a sufficiently wide single hidden layer can approximate any continuous function. However, in practice for image-like signals:

- Deep networks learn hierarchical, compositional representations more efficiently.
- Depth often needs fewer parameters than an equivalent shallow model for comparable expressivity.
- Optimization and generalization are typically better when intermediate features can be reused over multiple layers.

  From experiments:

- Pancake: 1.64M params, 77.35% validation accuracy.

- Tower: 0.54M params, 78.21% validation accuracy.
- Champion: 2.13M params, 79.16% validation accuracy.

This indicates that pure width is not sufficient; structured depth plus regularization improved validation performance.

D2. Confusion Analysis

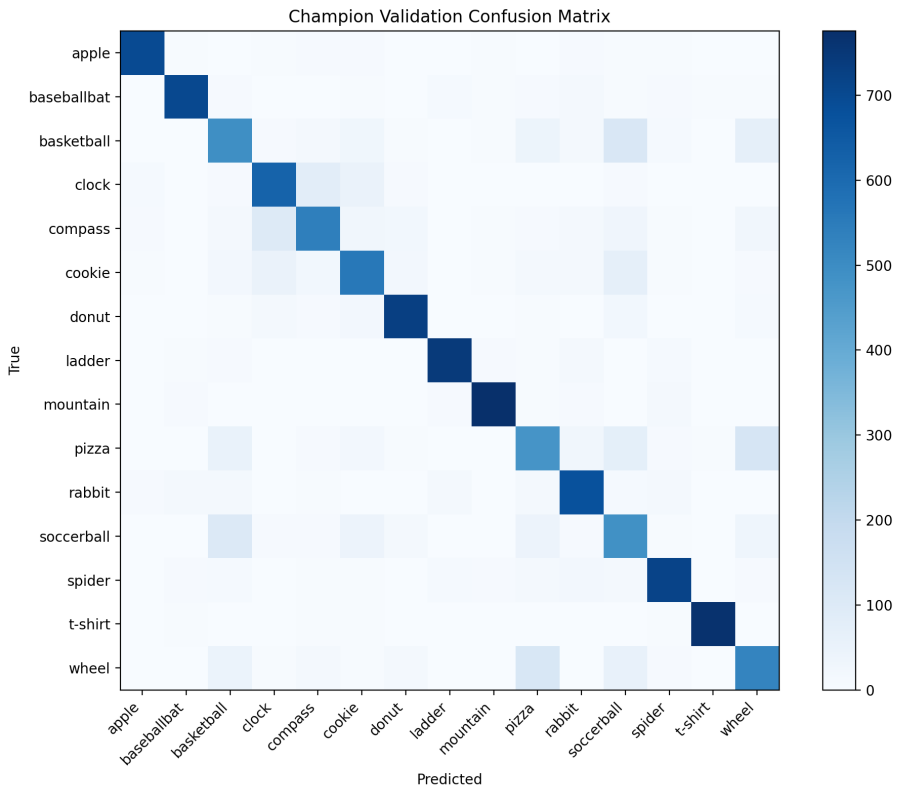
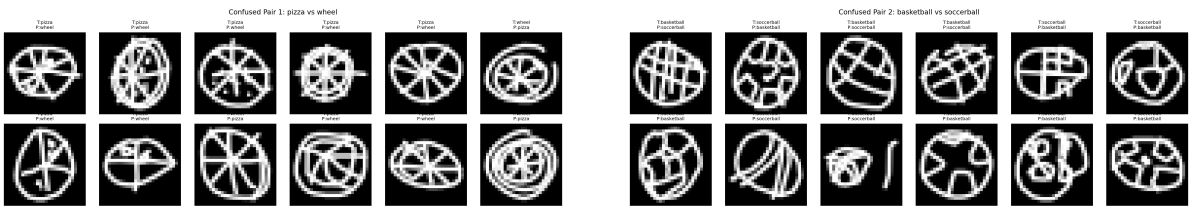


Figure 3: Confusion matrix of the selected champion model on local validation split.

Top-2 most confused class pairs:

1. **pizza vs wheel**: 250 total confusions (pizza→wheel: 129, wheel→pizza: 121)
2. **basketball vs soccerball**: 223 total confusions (basketball→soccerball: 117, soccerball→basketball: 106)



(a) pizza vs wheel (b) basketball vs soccerball

Figure 4: Visual examples from the two most confused class pairs.

Interpretation: A significant portion of these errors is due to *aleatoric uncertainty* in human doodles (high intra-class variation and ambiguous sketches), though some residual model limitations remain.

Part E: Architecture Rationale and Overfitting Control

Why this champion architecture?

The champion uses a balanced depth-width profile ([1024, 768, 512, 256]) to improve expressivity while staying under the 3M parameter cap. It combines:

- GELU activation for smoother nonlinear transitions.
- Batch normalization to stabilize deeper optimization.
- Dropout + weight decay to combat overfitting.

Overfitting behavior

Training curves show that training accuracy grows substantially higher than validation accuracy in later epochs, especially for wider models. This was mitigated using regularization and careful optimization choices.

Final numbers for leaderboard form

- **Model Parameters:** 2,125,071
- **Epochs used:** 34
- **Predictions file:** `submission.txt` (comma-separated, 15,000 predictions)

Reproducibility Artifacts

Key files included in the repository:

- Notebook: `<roll_num>_PA2.ipynb`
- Curves and analysis figures: `notebook_outputs/*.png`
- Experiment summaries: `notebook_outputs/model_summary.csv`, `top2_confusions.json`
- Champion checkpoint: `notebook_outputs/champion_best.pth`
- Leaderboard predictions: `submission.txt`

Generative AI Usage Disclosure

Prompt used (summary): Assistance was requested for structuring MLP experiments (Pancake/Tower/Champion), training pipeline design, confusion analysis workflow, and report drafting format.

Generated output used: Initial code/report structure suggestions and automation ideas.

Edits made by student: Dataset-specific adjustments, hyperparameter choices, architecture decisions, validation of outputs, and final report wording/organization were adapted for this assignment deliverable.

Note: The repository link above is the public submission repository for this assignment.